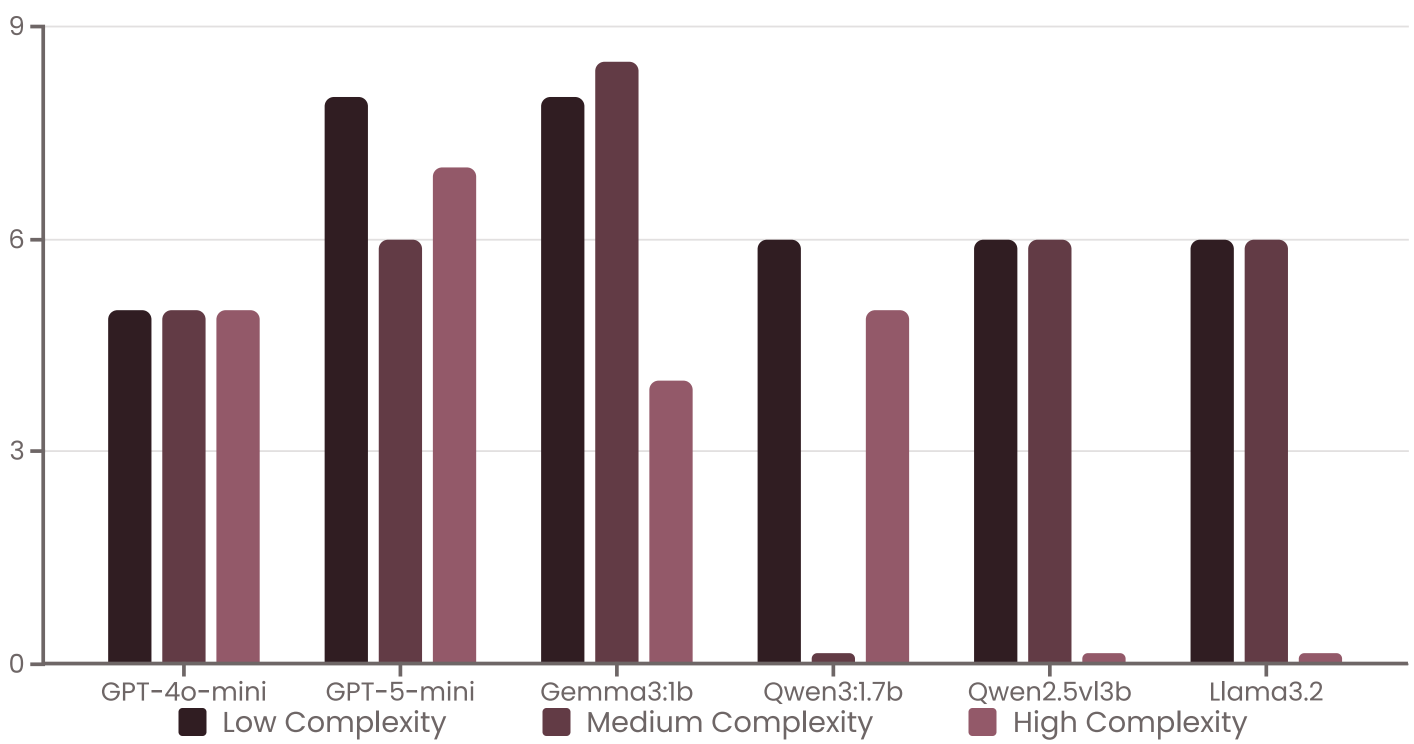
EXECUTIVE SUMMARY

Large Language Model Performance Benchmarking for Python code generation & Review Scenario

Six leading LLMs were rigorously tested against Euler problem sets spanning low, medium, and high complexity levels. Performance was measured using dual metrics: expert code review scores (1-10 scale) and static analysis quality through Pylint issue detection. Results reveal significant performance variance across model architectures and complexity thresholds.



Key Findings

- **Gemma3:1b** achieved highest medium-complexity score (8.5) but struggled with complex tasks
- **GPT-5-mini** demonstrated most consistent performance across all complexity tiers
- Three models failed to complete medium or high-complexity challenges
- Code quality inversely correlated with Pylint issues across all models

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LLMs Tested

Across three complexity tiers

50%

Failure Rate

On high-complexity tasks

8.5

Top Score

Gemma3 medium complexity

10

Max Pylint Issues

GPT-4o-mini high complexity